Technical Terminology & Definitions

Reference Guide for Electrical Team

NLC STEM Club - Electrical Systems Team

February 2026

1 TECHNICAL TERMINOLOGY & DEFINITIONS

- Fixed-Wing Environmental Monitoring UAV
- NLC STEM Club - Electrical Systems Team
- Reference Guide for Team Members

2 DOCUMENT PURPOSE:

- This glossary defines all technical terms, acronyms, and concepts used throughout
- the UAV electrical system documentation. Use this as a quick reference guide when
- encountering unfamiliar terminology.

3 ORGANIZATION:

- Terms are organized alphabetically within categories for easy lookup.

4 SECTION 1: ELECTRICAL TERMS & CONCEPTS

AMPERE (A) / MILLIAMPERE (mA)

- Unit of electrical current. 1 Ampere = 1000 milliamperes.
- Example: The flight controller draws 250mA (0.25A) of current.

AWG (American Wire Gauge)

- Standard system for measuring wire thickness. LOWER numbers = THICKER wire.
- 12 AWG: Very thick, high current (30A+) - motor wires
- 18 AWG: Medium, moderate current (5-10A) - power distribution
- 26 AWG: Thin, low current/signal (¡1A) - sensor wires
- The thicker the wire, the less resistance and more current it can safely carry.

5 BACKFEEDING

- Unwanted flow of power in the reverse direction through a circuit. In our system,
- this occurs if the ESC's +5V wire is connected to the FC, potentially damaging
- the FC's voltage regulator. SOLUTION: Cut the middle wire on ESC signal cable.

BEC (Battery Eliminator Circuit)

- A voltage regulator that converts high battery voltage (14.8V) down to 5V for
- electronics. "Eliminates" the need for a separate battery for radio equipment.
- Our ESC has a built-in 5V 3A BEC (Linear BEC - less efficient but simpler).

6 BROWNOUT

- Temporary voltage drop that causes electronics to malfunction or reset. Occurs
- when too much current is drawn from a power source, causing voltage to sag below
- minimum operating level. Our Dual-BEC design prevents FC brownout from servo loads.

CAPACITY (mAh or Ah)

- Milliamp-hours or Amp-hours. Measures how much charge a battery can store.
- Our battery: 2200mAh = can deliver 2200mA for 1 hour (or 1100mA for 2 hours)
- Higher capacity = longer flight time (but also heavier battery)

7 CONTINUITY

- Unbroken electrical path. Used in testing: "check continuity" means verify a
- wire is not broken and conducts electricity. Measured with multimeter in Ω mode.

8 CURRENT (I)

- Flow of electrical charge, measured in Amperes (A). Like water flow in a pipe.
- Formula: I = V / R (Current = Voltage / Resistance)
- Typical current: FC draws 250mA
- Peak current: Motor can draw 30A during full throttle

9 DUAL-BEC CONFIGURATION

9.1 Our chosen power architecture using TWO separate voltage regulators

- BEC 1 (ESC integrated): Powers flight-critical systems (FC, GPS, RX)
- BEC 2 (External UBEC): Powers high-current servos
- Advantage: Isolates servo power spikes from flight controller, preventing crashes.

10 DUTY CYCLE

- Percentage of time a signal is "ON" vs "OFF". Used in PWM control.
- 50% duty cycle = signal is high 50% of the time
- Servo control: 1ms pulse (5% duty cycle) = full left, 2ms (10%) = full right

ELECTRICAL NOISE / EMI (Electromagnetic Interference)

10.1 Unwanted electrical signals that interfere with sensitive electronics. Sources

- Motor/ESC switching (high-frequency noise)
- Power supply ripple
- Radio frequency interference
- Mitigation: Twisted wires, distance separation, filtering capacitors.

GND (Ground)

- Reference point for voltage measurement, typically 0V. Also completes circuit
- for current flow. CRITICAL: All components must share a common ground or signals
- will be corrupted. Our design uses star grounding topology.

LOAD

10.2 Any device that consumes electrical power. In our system

- Light load: GPS (60mA)
- Medium load: Arduino (263mA)
- Heavy load: Motor (10-30A)

11 OHMS LAW

- Fundamental relationship: V = I × R (Voltage = Current × Resistance)
- Also: I = V / R and R = V / I
- Used to calculate current draw, voltage drop, wire sizing, etc.

12 PARALLEL CONNECTION

- Components connected with all positive terminals together and all negative
- terminals together. Voltage stays same, currents add.
- Example: All 4 servos powered in parallel from UBEC (each gets 5V, currents sum).

13 POLARITY

- Direction of electrical flow. Positive (+) and Negative (-) terminals.

WARNING CRITICAL: Reversed polarity can instantly destroy electronics!

- Always verify RED = positive, BLACK = negative before connecting.

POWER (W, Watts)

- Rate of energy consumption. Formula: P = V × I (Power = Voltage × Current)
- Example: Motor at 14.8V drawing 20A = 296W power consumption
- Our battery: 2200mAh × 14.8V = 32.56Wh (Watt-hours) total energy

14 POWER RAIL

- Distribution bus that supplies power to multiple components at same voltage.

14.1 Our system has two 5V rails

- Rail 1: Flight-critical (FC, GPS, RX, Arduino) - ESC BEC
- Rail 2: Servo power (all 4 servos) - External UBEC

15 RESISTANCE (R)

- Opposition to current flow, measured in Ohms (Ω). Like friction in a pipe.
- Low resistance: Thick wires, good connections
- High resistance: Thin wires, corroded connections
- Causes voltage drop and heat generation.

16 SERIES CONNECTION

- Components connected end-to-end so current flows through each in sequence.
- Voltages add, current stays same.
- Example: Battery cells in series (3.7V + 3.7V + 3.7V + 3.7V = 14.8V for 4S).

17 SHORT CIRCUIT

- Unintended direct connection between positive and negative, bypassing load.
- Results in VERY high current, heat, fire risk, component damage.

WARNING ALWAYS use fuse/protection and inspect for bare wire touching!

18 STAR GROUNDING

- Grounding topology where all ground wires connect to ONE central point (battery
- negative) rather than daisy-chaining. Prevents ground loops and voltage differences
- between components. REQUIRED for our system's proper operation.

UBEC (Universal Battery Eliminator Circuit)

- Advanced switching voltage regulator (vs linear BEC). More efficient, handles
- higher input voltage range, generates less heat. Our external UBEC is 5V 6A
- switching regulator for servo power.

19 VOLTAGE (V)

- Electrical potential difference, measured in Volts. Like water pressure in a pipe.
- Battery voltage: 14.8V nominal (13.2-16.8V range for 4S LiPo)
- Regulated voltage: 5.0V for electronics
- Signal voltage: 3.3V or 5V logic levels

20 VOLTAGE DROP

- Reduction in voltage due to resistance in wires/connections. Formula: V_drop = I × R
- Problem: Long thin wires carrying high current have significant voltage drop.
- Solution: Use appropriately thick wire for current level (AWG sizing).

21 VOLTAGE SAG

- Temporary voltage drop when high current is suddenly drawn. Battery internal
- resistance causes voltage to "sag" under heavy load. Can cause brownout if
- excessive. Our high C-rating battery (130C) minimizes voltage sag.

22 SECTION 2: COMPONENT TERMINOLOGY

23 ARDUINO

- Open-source microcontroller platform. Our Arduino Nano uses ATmega328P chip
- (8-bit, 16MHz, 32KB flash) for data logging and sensor interfacing. Programmed
- in C/C++ using Arduino IDE.

24 BAROMETER

- Atmospheric pressure sensor (BMP280 on flight controller). Measures air pressure
- to calculate altitude. Used for altitude hold and vertical navigation.

BLDC MOTOR (Brushless DC Motor)

- Electric motor with electronic commutation (no brushes). More efficient, reliable,

- and powerful than brushed motors. Requires ESC for speed control. Our D2830EVO
- is a 3-phase outrunner BLDC motor.

C-RATING (Battery)

- Discharge rate capability of LiPo battery. Formula: Max Current = Capacity × C-Rating
- Our battery: 2.2Ah × 130C = 286A maximum burst current (never approach this!)
- Higher C-rating = lower internal resistance = less voltage sag under load.

COMPASS (Magnetometer)

- Sensor that detects Earth's magnetic field for heading determination. Integrated
- in BN-220 GPS module (HMC5883L chip). Requires calibration and distance from
- magnetic interference (motors, current-carrying wires).

ESC (Electronic Speed Controller)

- Power electronics that control brushless motor speed. Converts DC battery power
- to 3-phase AC for motor. Receives PWM throttle signal from FC, outputs variable
- frequency/amplitude to motor. Our 40A ESC has integrated 5V BEC.

FC (Flight Controller)

- Central computer that stabilizes aircraft and executes flight commands. Our
- MATEK F405-WMN has:
- STM32F405 processor (32-bit ARM, 168MHz)
- IMU (gyro + accelerometer) for attitude sensing
- Runs iNav or ArduPilot firmware

25 FOLDING PROPELLER

- Propeller with blades that fold back when motor is off (reduces drag in glide).
- Spring-loaded to deploy under centrifugal force when spinning. Used on our
- motor for efficient glide performance.

GPS (Global Positioning System)

- Satellite navigation system. Our BN-220 module receives signals from GPS
- satellites to determine 3D position (latitude, longitude, altitude) and velocity.
- Requires clear view of sky; 6+ satellites for good fix.

26 GYROSCOPE

- Sensor that measures angular rotation rate (degrees per second). Part of IMU
- on flight controller. Used for stabilization - detects when aircraft is rotating
- and commands control surfaces to counteract.

27 HEAT SHRINK TUBING

- Plastic tubing that shrinks when heated, used to insulate electrical connections.
- Always cover solder joints and bare wire with heat shrink to prevent shorts.

IMU (Inertial Measurement Unit)

- Combined sensor package: 3-axis gyroscope + 3-axis accelerometer = 6 DOF
- (degrees of freedom). Measures rotation and acceleration in all directions.
- Flight controller uses IMU data for stabilization and attitude estimation.

28 JST CONNECTOR

28.1 Small standardized connector series. Common types

- JST-XH: Battery balance connector
- JST-SH: 1.0mm pitch (GPS, receiver)
- JST-PH: 2.0mm pitch (general electronics)

KV RATING (Motor)

- RPM per volt constant. Our 1000KV motor spins at 1000 RPM for every volt applied.
- At 14.8V: 14,800 RPM (no load)
- Lower KV = more torque, larger props; Higher KV = less torque, smaller props.

LiPo (Lithium Polymer Battery)

- Rechargeable battery with high energy density. Chemistry: Lithium-ion with
- polymer electrolyte. Nominal voltage: 3.7V per cell.
- 1S = 3.7V, 2S = 7.4V, 3S = 11.1V, 4S = 14.8V (our battery)

WARNING DANGEROUS: Can catch fire if damaged, overcharged, or short-circuited!

OSD (On-Screen Display)

- Overlays flight telemetry on FPV video feed. MATEK F405-WMN has integrated OSD
- chip that can display battery voltage, current, altitude, GPS data, etc. on
- screen. Configured in iNav/ArduPilot.

29 OUTRUNNER MOTOR

- BLDC motor where outer case rotates and center is stationary (opposite of
- inrunner). Higher torque, better for direct-drive propellers. Our D2830EVO is
- an outrunner.

30 RECEIVER (RX)

- Radio receiver that picks up transmitter signals. Our HAPPYMODEL EP2 receives
- ExpressLRS 2.4GHz radio commands and sends them to FC via CRSF or SBUS protocol.

31 SENSOR FUSION

- Algorithm that combines data from multiple sensors (GPS, IMU, barometer, compass)
- to get better position/attitude estimate than any single sensor. Flight controller
- firmware does this automatically.

SERVO

- Electromechanical actuator with position feedback. Receives PWM pulse (1-2ms)
- and rotates output shaft to corresponding angle. Our ES08MA II servos have:
- Metal gears (durable)
- 180° rotation
- Proportional control

32 TRANSMITTER (TX)

- Hand-held radio controller. Pilot inputs (sticks, switches) converted to radio
- signals and sent to receiver on aircraft. Can also refer to data transmit pin
- on serial connection.

UBEC (Previously Defined)

- See Section 1. External voltage regulator providing 5V @ 6A for servo power.

33 XT60 CONNECTOR

- High-current connector rated for 60A continuous. Yellow plastic housing with
- spring-pin contacts. Standard for 3S-6S LiPo batteries. Gender: Male on battery,
- Female on ESC/cables.

34 SECTION 3: PROTOCOLS & COMMUNICATION STANDARDS

35 ANALOG SIGNAL

- Continuously variable voltage representing data. Our MQ135 gas sensor outputs
- 0-5V analog voltage proportional to gas concentration. Read by Arduino ADC
- (Analog-to-Digital Converter).

36 BAUD RATE

36.1 Speed of serial communication in bits per second (bps). Common rates

- 9600 baud: GPS default (slow but reliable)
- 115200 baud: Fast serial (firmware upload, debugging)
- 420000 baud: CRSF protocol from ELRS receiver
- Both devices must use SAME baud rate to communicate.

CRSF (Crossfire Protocol)

- Bi-directional serial protocol for RC control and telemetry. Developed by TBS,
- now used by ExpressLRS. 420000 baud, low latency, supports up to 16 channels
- plus telemetry data. Our ELRS receiver uses CRSF to talk to FC.

37 DIGITAL SIGNAL

- Discrete on/off voltage levels representing binary data (0 or 1).
- Logic LOW: 0V (binary 0)
- Logic HIGH: 3.3V or 5V (binary 1)
- Used for: I2C, SPI, UART, PWM control signals.

ELRS (ExpressLRS)

37.1 Open-source long-range radio control system using 2.4GHz or 900MHz. Features

- Very low latency (5-20ms)
- Long range (10+ km)
- Frequency hopping for interference immunity
- Over-the-air updates

- Our EP2 receiver uses ELRS protocol.

I2C (Inter-Integrated Circuit)

- Two-wire serial protocol for short-distance communication. Uses SDA (data) and
- SCL (clock) lines. Multiple devices on same bus (each has unique address).
- Our BME280 sensor uses I2C to send temperature/humidity/pressure data to Arduino.

NMEA (National Marine Electronics Association)

37.2 Text-based protocol for GPS data. Human-readable sentences like

38 $GPGGA,123519,4807.038,N,01131.000,E,1,08,0.9,545.4,M,46.9,M,,*47

- Contains position, time, satellites, etc. Alternative to binary UBX protocol.

PWM (Pulse Width Modulation)

38.1 Signal where information is encoded in pulse duration. Used for

- Servo control: 1ms = -90°, 1.5ms = center, 2ms = +90°
- Motor throttle: 1ms = 0%, 2ms = 100%
- Frequency: typically 50Hz (20ms period) for servos
- Analog value represented by digital signal duty cycle.

SBUS (Serial Bus)

- Inverted serial protocol from FrSky. 100000 baud, 25ms update, 16 channels.
- Less efficient than CRSF but widely compatible. Fallback option if CRSF doesn't
- work with our FC firmware.

SPI (Serial Peripheral Interface)

- Four-wire serial protocol: MOSI (master out), MISO (master in), SCK (clock),
- CS (chip select). Fast, full-duplex communication. Our MicroSD card module uses
- SPI to transfer data at high speed from Arduino.

39 TELEMETRY

39.1 Real-time transmission of aircraft data back to ground station. Can include

- battery voltage, current, altitude, GPS position, speed, RSSI (signal strength).
- CRSF protocol supports bi-directional telemetry.

UART (Universal Asynchronous Receiver-Transmitter)

- Serial communication protocol using TX (transmit) and RX (receive) wires.
- Asynchronous = no clock signal, must agree on baud rate.
- UART1 on FC: Connected to GPS (9600 baud)
- UART2 on FC: Connected to receiver (420000 baud)

40 UBX PROTOCOL

- Binary GPS protocol from u-blox. More efficient than NMEA, higher update rate,
- configurable messages. Our BN-220 GPS supports both UBX and NMEA.

41 SECTION 4: AVIATION & UAV TERMINOLOGY

42 AILERON

- Control surface on wing trailing edge that controls roll (banking). Differential
- movement: left aileron up + right aileron down = roll left.
- Our aircraft has 2 ailerons (Servo 1 and Servo 2).

43 AUTONOMOUS FLIGHT

- Aircraft flies pre-programmed mission without pilot input. Uses GPS waypoints,
- altitude commands, loiter circles. FC firmware executes mission autonomously.
- Requires GPS lock and properly tuned flight controller.

44 BLACKBOX

- Flight data recorder that logs sensor data at high rate (1000Hz+) for post-
- flight analysis. Can be onboard flash memory or microSD card. Used for tuning
- PIDs, diagnosing crashes, analyzing performance.

45 CONTROL SURFACE

45.1 Movable aerodynamic surface that controls aircraft attitude

- Ailerons: Roll control
- Elevator: Pitch control
- Rudder: Yaw control

46 ELEVATOR

- Control surface on horizontal tail that controls pitch (nose up/down).
- Our Servo 3 controls the elevator.

ELRS (Already Defined)

- See Section 3 - ExpressLRS long-range control system.

47 FAILSAFE

47.1 Pre-programmed behavior when radio link is lost. Options

- Hold position (loiter)
- Return to home (RTH) and land
- Immediate land
- Cut throttle (glide down)
- Configured in flight controller firmware.

48 FIXED-WING

- Aircraft with continuous rigid wings (like airplane), as opposed to rotorcraft
- (quadcopter). Our UAV is fixed-wing design. Requires forward motion for lift;
- can glide efficiently but cannot hover.

FPV (First Person View)

- Flying from onboard camera perspective. Video transmitted to pilot's goggles/
- screen in real-time. Our project focuses on autonomous flight, not FPV, but
- FC has OSD capability for future FPV addition.

49 GPS LOCK / FIX

- Successful acquisition of GPS satellite signals allowing position determination.

49.1 Requires

- Clear sky view
- 4+ satellites for 2D fix (lat/long)
- 6+ satellites for 3D fix (lat/long/altitude)
- Indicated by LED on GPS module.

IMU (Already Defined)

- See Section 2 - Inertial Measurement Unit for stabilization.

50 LOITER / POSITION HOLD

- Flight mode where aircraft maintains fixed GPS position (circles within radius).
- Requires GPS lock. Useful for observing area or waiting for telemetry.

PID (Proportional-Integral-Derivative)

- Control algorithm used for stabilization. FC continuously adjusts control surfaces
- to maintain desired attitude:
- P: Correction proportional to error (immediate response)
- I: Correction based on accumulated error (eliminates steady-state error)
- D: Correction based on rate of error change (dampening)
- Tuning PID values is critical for stable flight.

PITCH

- Rotation about lateral axis (nose up/down). Controlled by elevator.

51 PROPELLER

51.1 Rotating blade that converts motor power to thrust. Specifications

- Diameter × Pitch: 1060 = 10” diameter, 6” pitch
- Pitch = distance propeller advances in one rotation
- Folding prop: Blades fold back when not spinning

ROLL

- Rotation about longitudinal axis (banking left/right). Controlled by ailerons.

RTH (Return to Home)

- Autonomous flight mode where aircraft flies back to takeoff location using GPS.
- Can be triggered by pilot command, failsafe, or low battery.

52 RUDDER

- Control surface on vertical tail that controls yaw (nose left/right).
- Our Servo 4 controls the rudder. Less critical in calm wind than ailerons/elevator.

53 STABILIZATION

- Automatic correction to maintain level flight. Gyros detect rotation, FC commands
- control surfaces to counteract. Modes:
- Acro: No stabilization (manual input only)
- Angle: Self-leveling (returns to level when sticks centered)
- Horizon: mix of Acro and Angle

UAV (Unmanned Aerial Vehicle)

- Aircraft without human pilot onboard. Controlled remotely or autonomously.
- Synonyms: drone, UAS (Unmanned Aircraft System), remotely piloted aircraft.

YAW

- Rotation about vertical axis (nose left/right). Controlled by rudder.

54 SECTION 5: SENSOR & MEASUREMENT TERMINOLOGY

ADC (Analog-to-Digital Converter)

- Circuit that converts analog voltage to digital number. Arduino Nano has 10-bit
- ADC (0-1023 range) for reading analog sensors. MQ135 gas sensor output is read
- by ADC on pin A0.

55 BME280

55.1 Combined environmental sensor IC (Bosch Sensortec)

- Barometric pressure: 300-1100 hPa (altitude 0-30,000 ft)
- Temperature: -40 to +85°C (±1°C accuracy)
- Humidity: 0-100% RH (±3% accuracy)
- Uses I2C interface. Perfect for weather monitoring mission.

56 BURN-IN PERIOD

- Initial operation time required for sensor to stabilize. MQ135 gas sensor requires
- 24-48 hours of continuous power before accurate readings. Tin oxide sensor
- element needs to heat up and stabilize chemically.

57 CALIBRATION

57.1 Procedure to adjust sensor readings to known reference values. Examples

- Compass: Rotate aircraft through 360° to map magnetic field distortions
- ESC: Set throttle range endpoints (min/max PWM values)
- Gas sensor: Expose to fresh air and record baseline reading

CSV (Comma-Separated Values)

- Simple text file format for data logging. Each line is one record, values
- separated by commas. Example:
- Time,Temp,Humidity,AirQuality
- 0,24.5,65,450
- Our Arduino saves sensor data in CSV format for analysis in Excel/Python.

58 DATA LOGGING

58.1 Recording sensor measurements over time for post-flight analysis. Our system

- Timestamp from millis() function
- BME280 readings every second
- MQ135 readings every second
- Written to microSD card in CSV format

59 I2C ADDRESS

59.1 Unique 7-bit number identifying device on I2C bus. BME280 default address

- 0x76 or 0x77 (solder jumper selectable). Scanner sketch can find device addresses.

MicroSD CARD

- Small removable flash storage. Our module uses SPI interface for data transfer.
- Format: FAT32 recommended (up to 32GB cards). Stores sensor data logs as CSV files.

MQ135

59.2 Metal oxide gas sensor for air quality monitoring. Detects

- CO2 (carbon dioxide)
- NH3 (ammonia)
- NOx (nitrogen oxides)
- Alcohol, smoke, benzene
- Outputs analog voltage (0-5V) proportional to gas concentration. Requires 5V
- supply and 24-48 hour burn-in period. Not highly selective (detects multiple gases).

60 RESOLUTION

60.1 Smallest change a sensor can detect. Examples

- BME280 temperature: 0.01°C resolution
- Arduino ADC: 5V / 1024 = 4.88mV resolution (10-bit)
- Higher resolution = more precise measurements.

61 SAMPLING RATE

- Frequency of sensor measurements. Our system: 1 Hz (one reading per second).
- Higher sampling rate = more data, faster battery drain, larger log files.

SENSOR FUSION (Already Defined)

- See Section 3 - Combining multiple sensor inputs for better accuracy.

62 TIMESTAMP

- Time marker for each data point. Arduino uses millis() function (milliseconds
- since power-on). Format: Unix timestamp or formatted date/time string.

63 SECTION 6: SOFTWARE & FIRMWARE TERMINOLOGY

64 ARDUINO IDE

- Software development environment for programming Arduino boards. Uses simplified
- C/C++ language. Includes code editor, compiler, and upload tool. Free and
- open-source.

ArduPilot

- Open-source autopilot software for fixed-wing, multirotor, and other vehicles.
- Features: GPS navigation, autonomous missions, advanced sensors. Alternative
- firmware for our MATEK F405-WMN (in addition to iNav).

65 BOOTLOADER

- Small program that allows firmware updates via USB. Pre-installed on Arduino
- and flight controllers. Runs first on power-up, checks for upload request.

66 CONFIGURATOR

- GUI software for flight controller setup. Options:
- iNav Configurator (for iNav firmware)
- Mission Planner (for ArduPilot)
- Betaflight Configurator (for Betaflight)
- Used to configure PID values, modes, calibration, OSD layout.

67 FIRMWARE

- Low-level software running directly on microcontroller. Read-only memory,
- flashed via USB. Examples:
- iNav on flight controller
- Arduino sketch on Arduino Nano
- ELRS on receiver

68 FLASHING

68.1 Uploading firmware to microcontroller's flash memory. Process

- 1. Connect device via USB
- 2. Select COM port in software
- 3. Click "Flash" button
- 4. Wait for completion (device reboots automatically)

69 GITHUB

- Online platform for hosting open-source code. ELRS, iNav, ArduPilot all hosted
- on GitHub. Can download latest versions, report bugs, contribute code.

iNav

- Open-source flight control firmware optimized for fixed-wing and long-range
- flight. Features GPS navigation, missions, return-to-home. Alternative to
- ArduPilot, easier setup. Recommended for our project.

70 LIBRARY

- Pre-written code that provides functionality for common tasks. Arduino libraries
- for our project:
- Adafruit_BME280: BME280 sensor interface
- SD: MicroSD card file operations
- Wire: I2C communication

71 MISSION PLANNING

71.1 Creating pre-programmed flight path with GPS waypoints. Software

- Mission Planner (ArduPilot)
- iNav Configurator (iNav)
- Define takeoff, waypoints (lat/long/alt), loiter circles, landing.

72 OSD LAYOUT

- Configuration of on-screen display elements (position, size, visibility).
- Can show: battery voltage, current, altitude, GPS position, speed, heading,
- artificial horizon, etc.

73 SKETCH

- Arduino program file (.ino extension). Contains setup() function (runs once)
- and loop() function (runs repeatedly). Our data logging sketch reads sensors
- and writes to SD card.

USB BOOTLOADER (See BOOTLOADER)

74 WAYPOINT

- GPS coordinate (latitude, longitude, altitude) defining point in mission.
- Aircraft flies from waypoint to waypoint autonomously. Can specify actions
- at each waypoint (loiter time, heading, speed).

75 SECTION 7: SAFETY & TESTING TERMINOLOGY

76 BALANCE CHARGING

- LiPo charging method that ensures all cells reach same voltage. Uses balance
- connector (JST-XH) to monitor individual cell voltages. REQUIRED for safe
- LiPo charging. Never charge without balance lead connected!

77 BENCH TEST

77.1 Testing on workbench before installation in aircraft. Advantages

- Safer (no propeller)
- Easier troubleshooting (access to all components)
- Can measure voltages and currents
- Always bench test before flight test!

78 CELL VOLTAGE

78.1 Voltage of individual cell in multi-cell battery. LiPo safe voltages

- Full charge: 4.2V per cell
- Nominal: 3.7V per cell
- Safe minimum: 3.3V per cell
- DANGER: 3.0V per cell (damages cell permanently)
- 4S battery: 16.8V full, 14.8V nominal, 13.2V minimum

79 CONTINUITY TEST

- Using multimeter to verify electrical connection (wire not broken). Set meter
- to Ω mode, touch probes to both ends of wire. Should beep/read ˜0Ω if continuous.

80 FAILSAFE TEST

80.1 Verifying aircraft behavior when radio link is lost. Process

- 1. Arm aircraft on ground
- 2. Turn off transmitter
- 3. Verify failsafe activates (motor cuts, servos go to preset positions)
- 4. Turn transmitter back on
- 5. Verify control restored

81 FIRE EXTINGUISHER

- Required safety equipment for LiPo batteries. Class D (metal fire) or ABC
- extinguisher. LiPo fires extremely hot and dangerous. Alternative: Sand bucket
- or metal container to smother fire.

FUSE

- Protective device that breaks circuit if current exceeds rated value. Should
- be installed on battery positive lead. Our system: 40A fuse protects against
- short circuits, prevents battery damage and fire risk.

82 GROUND TEST

82.1 Final system check before flight

- Power on sequence (battery last)
- GPS lock verification
- Radio range test (walk 50m away, verify control)
- Failsafe test
- Control surface deflection check (full travel, correct direction)

83 LIPO-SAFE BAG

- Fire-resistant bag for charging and storing LiPo batteries. Contains fire if
- battery catches fire during charging. NOT fireproof (will eventually burn through),
- but provides time to move outside. Alternative: Metal ammo can.

84 PUFFED BATTERY

- Damaged LiPo battery with swollen cells (bulging). Caused by overcharge, over-
- discharge, physical damage, or age. [WARNING] EXTREMELY DANGEROUS - can explode or
- catch fire! Dispose immediately at battery recycling center.

85 RANGE TEST

- Verifying radio control range before flight. ELRS: Built-in range test mode
- reduces TX power to simulate long distance. Walk away from aircraft until warning
- (should be ¿100m). Verify control and failsafe work correctly.

86 SMOKE TEST

- Informal term: First power-on of new build. "If it doesn't smoke, it works!"

86.1 Serious first power-up procedure

- 1. Visual inspection (no shorts, correct polarity)
- 2. Connect battery slowly, watch for sparks
- 3. Feel components for excessive heat
- 4. Measure voltages with multimeter
- 5. If anything smokes/smells: DISCONNECT IMMEDIATELY!

87 STORAGE VOLTAGE

- Safe voltage for long-term LiPo storage: 3.8V per cell (15.2V for 4S).
- Reduces stress on cells, prolongs battery life. Use charger's "storage mode"
- if not flying for ¿1 week.

88 THERMAL RUNWAY

- Uncontrolled exothermic reaction in damaged LiPo battery. Cell overheats, causing
- chemical breakdown, generating more heat, causing neighboring cells to fail.
- Results in fire/explosion. Prevention: Proper charging, damage inspection,
- voltage monitoring.

89 VOLTAGE ALARM

- Small device that beeps when LiPo voltage drops below set threshold. Plugs into
- balance connector. Prevents over-discharge during flight. Set alarm at 3.5V/cell
- (14.0V for 4S) for safety margin.

90 SECTION 8: CONNECTORS & MECHANICAL TERMINOLOGY

91 BULLET CONNECTOR

91.1 Cylindrical push-fit electrical connector. Common sizes

- 2mm: Low current signal wires
- 3.5mm: Motor phase wires (our ESC to motor)
- 4mm: High current battery connections
- Male = pin (on wire), Female = socket (on wire or component)

92 CRIMPING

- Mechanical joining of connector to wire using special tool. Alternative to
- soldering for some connectors (JST, Dupont). Requires proper crimp tool for
- connector size.

93 DUPONT CONNECTOR

- 0.1” pitch plastic housing with crimped pins. Used for breadboard connections,
- Arduino headers, sensor wiring. Can make custom cable lengths. Separate male
- and female pins.

HEAT SHRINK (Already Defined)

- See Section 1 - Insulating tubing that shrinks when heated.

JST CONNECTOR (Already Defined)

- See Section 2 - Small standardized connector family.

94 MOUNTING TAPE / FOAM TAPE

- Double-sided foam tape for securing electronics. Provides vibration dampening
- and electrical insulation. Use for mounting FC, GPS, receiver.

95 SERVO ARM / SERVO HORN

- Plastic attachment on servo output shaft. Connects to pushrod/linkage for
- control surface actuation. Usually cross-shaped with multiple holes for different
- throw distances.

96 SERVO EXTENSION

- Cable to extend servo wire length. Standard 3-pin, male-female. Available in
- various lengths (6", 12", 24"). Allows mounting servos far from FC.

97 SOLDER / SOLDERING

97.1 Joining metal parts with molten metal alloy (tin-lead or lead-free). Process

- 1. Heat joint with soldering iron (300-350°C)
- 2. Apply solder (melts and flows into joint)
- 3. Remove iron, let cool (2-3 seconds)
- Good solder joint: Shiny, smooth, no cold spots. Bad joint: Dull, cracked,
- blobby (redo it).

98 TWIST TIE / ZIP TIE / CABLE TIE

- Plastic strap for bundling and securing wires. Prevents wires from snagging
- on moving parts, keeps electronics neat. Use locking zip ties (not reusable)
- for permanent installation.

99 WIRE STRIPPING

- Removing insulation from wire end to expose conductor. Use wire strippers
- (adjustable for gauge) or knife (carefully). Strip enough for connection but
- not excessive (keeps joint mechanically strong).

XT30 / XT60 (XT60 Already Defined)

99.1 High-current connector series

- XT30: 30A rated, smaller/lighter (good for UBEC power input)
- XT60: 60A rated, standard for 3S-6S batteries (our main battery connector)
- XT90: 90A rated, heavy duty (overkill for our project)

100 SECTION 9: TROUBLESHOOTING TERMS

101 COLD SOLDER JOINT

- Poor solder connection that looks dull/grainy instead of shiny. Caused by moving
- joint before solder cools, or insufficient heat. Results in intermittent connection.
- Fix: Reheat with soldering iron until shiny and smooth.

102 GROUND LOOP

- Unwanted current flow between multiple ground paths at different voltages.
- Causes noise, erratic behavior, flickering OSD. Prevention: Star grounding
- topology (all grounds to ONE point).

103 INTERMITTENT CONNECTION

103.1 Electrical connection that works sometimes but not always. Causes

- Cold solder joint
- Loose connector
- Broken wire strand (partially broken)
- Corrosion
- Troubleshooting: Wiggle wires while powered to reproduce problem.

104 MAGIC SMOKE

- Humorous term for electronics component failure. "I let the magic smoke out =
- I broke it." Expensive lesson - always double-check polarity and connections!

105 NO GPS FIX / NO GPS LOCK

- GPS module cannot determine position. Troubleshooting:
- Verify clear sky view (no roof, no dense trees)
- Check antenna not damaged/disconnected
- Verify TX/RX wires not swapped
- Wait longer (cold start can take 5+ minutes)
- Check baud rate matches FC configuration

106 OSCILLATION

- Unstable rapid shaking or vibration. In flight: PID values too high (reduce P
- and D gains). In servo: Mechanical binding or damaged feedback potentiometer.

107 POWER CYCLING

- Turning device off and on again. Basic troubleshooting step that clears transient
- errors, resets firmware. For our system: Disconnect battery, wait 5 seconds,
- reconnect.

108 REVERSED SERVO

- Servo moves opposite direction from command (up command = down movement). Fix
- in FC configurator: Enable "reverse" checkbox for that channel. Do NOT physically
- reverse servo wires (changes null position).

109 RX LOSS / LINK LOST

109.1 Radio receiver not receiving transmitter signal. Causes

- Transmitter off or too far away
- Receiver antenna damaged
- Frequency interference
- Low transmission power
- Result: Failsafe activates. Check RSSI (signal strength) telemetry.

SHORT CIRCUIT (Already Defined)

- See Section 1 - Direct connection between + and -, bypassing load.

110 VOLTAGE SPIKE

- Brief surge in voltage above normal level. Caused by inductive kickback (motor
- stopping), switching transients. Can damage sensitive electronics. Mitigation:
- Capacitors across power supply, TVS diodes on signal lines.

111 SECTION 10: ACRONYMS QUICK REFERENCE

ADC Analog-to-Digital Converter
AHRS Attitude and Heading Reference System
ARM Advanced RISC Machine (processor architecture)
AWG American Wire Gauge
BEC Battery Eliminator Circuit
BLDC Brushless DC Motor
CRSF Crossfire Protocol
CSV Comma-Separated Values
DC Direct Current (as opposed to AC - Alternating Current)
DOF Degrees of Freedom
ELRS ExpressLRS (radio control protocol)
EMI Electromagnetic Interference
ESC Electronic Speed Controller
FC Flight Controller
FPV First Person View
GND Ground (0V reference)
GPS Global Positioning System
GUI Graphical User Interface
I2C Inter-Integrated Circuit
IDE Integrated Development Environment
IMU Inertial Measurement Unit
JST Japan Solderless Terminal (connector type)
KV RPM per Volt constant (motor rating)
LED Light Emitting Diode
LiPo Lithium Polymer (battery type)
mAh Milliamp-hours (battery capacity)
MCU Microcontroller Unit
NMEA National Marine Electronics Association
OSD On-Screen Display
PCB Printed Circuit Board
PID Proportional-Integral-Derivative (control algorithm)
PPM Pulse Position Modulation
PWM Pulse Width Modulation
RC Radio Control
RF Radio Frequency
RSSI Received Signal Strength Indicator
RTH Return to Home
RX Receiver (or Receive pin on serial)
SBUS Serial Bus (FrSky protocol)
SCL Serial Clock (I2C)
SDA Serial Data (I2C)
SPI Serial Peripheral Interface
TX Transmitter (or Transmit pin on serial)
UAV Unmanned Aerial Vehicle
UART Universal Asynchronous Receiver-Transmitter
UBEC Universal Battery Eliminator Circuit

UBX u-blox binary GPS protocol
USB Universal Serial Bus
VOC Volatile Organic Compounds (what MQ135 detects)

112 APPENDIX: COMMON FORMULAS

113 OHM'S LAW:

- V = I × R (Voltage = Current × Resistance)
- I = V / R (Current = Voltage / Resistance)
- R = V / I (Resistance = Voltage / Current)

114 POWER:

- P = V × I (Power = Voltage × Current)
- $P = I^2 \times R$ (Power = Current squared × Resistance)
- $P = V^2 / R$ (Power = Voltage squared / Resistance)

115 BATTERY CAPACITY:

- Runtime (hours) = Capacity (mAh) / Current (mA)
- Energy (Wh) = Capacity (Ah) × Voltage (V)

WIRE CURRENT CAPACITY (Approximate):

116 12 AWG: 20-25A

117 14 AWG: 15-20A

118 16 AWG: 10-15A

119 18 AWG: 7-10A

120 20 AWG: 5-7A

121 22 AWG: 3-5A

122 24 AWG: 2-3A

- 26 AWG: 1-2A (signal wires only)

123 SERVO TRAVEL:

- Travel (degrees) = (Pulse Width - 1.5ms) × 90° per 0.5ms
- Center: 1.5ms = 0°
- Min: 1.0ms = -90°
- Max: 2.0ms = +90°

124 LIPO CELL VOLTAGES:

- Full charge: 4.20V per cell (16.8V for 4S)
- Nominal: 3.70V per cell (14.8V for 4S)
- Storage: 3.80V per cell (15.2V for 4S)
- Low warning: 3.50V per cell (14.0V for 4S)
- Critical min: 3.30V per cell (13.2V for 4S)
- Damaged: ¡3.00V per cell (DO NOT USE!)

125 END OF TERMINOLOGY REFERENCE

This document should be used as a quick reference guide throughout the project.

When you encounter an unfamiliar term in meeting materials, wiring diagrams, or troubleshooting, refer to this glossary.

Recommended: Print this document and keep it accessible during build sessions.

125.1 For questions not answered here, consult

- Arduino reference: arduino.cc
- iNav documentation: github.com/iNavFlight/inav/wiki
- ELRS documentation: expresslrs.org
- Component datasheets (available on manufacturer websites)

Good luck with your build!